Chapter 15 Notes

Section 0

Exercise 0.1

I will note that I was not able to prove this unless we assume that A has some cardinality (or rather, that A has a choice function). The parts (a)-(c) of this exercise basically follows notes I had in the "Theorem 1.1" of Section 1 of the Chapter 11 notes.

(d) We show by transfinite induction on all infinite ordinals that for each infinite ordinal α, the set L_α has cardinality at most $\overline{\overline{\alpha}}$.

For the successor case, suppose the result holds for some L_α. If $\alpha+1$ is infinite, then α must also be infinite. By part (c) of the exercise, it follows that $\overline{\overline{L_{\alpha+1}}} = \overline{\overline{L_\alpha}}$, which completes the proof in this case.

For the limit case, suppose that for all ordinals β less than some limit ordinal α, $\overline{\overline{L_\beta}} \leq \beta$. Because α is a limit ordinal, $L_\alpha = \cup_{\beta<\alpha} L_\beta$. If $L_\alpha = L_\omega$, we are done, as L_ω is denumerable. Otherwise, it is easy to show that all L_β for $\beta < \alpha$ are such that $\overline{\overline{L_\beta}} \leq \overline{\overline{\alpha}}$. Because $L_\alpha = \cup S$ where S is the set of all L_β with $\beta < \alpha$, it follows that $\overline{\overline{L_\alpha}} \leq \overline{\overline{\alpha}}$ (because each memeber of S has at most cardinality $\overline{\overline{\alpha}}$, and there are α members of S).

(e) For this problem, note that if α is an infinite ordinal, the number of successor ordinals in α must be $\overline{\overline{\alpha}}$ (Reason: It cannot be that the number of successor ordinals in α was less than $\overline{\overline{\alpha}}$ and the number of limit ordinals in α was less than $\overline{\overline{\alpha}}$, or else the number of elements in α would be less than $\overline{\overline{\alpha}}$. If the number of limit ordinals in α was $\overline{\overline{\alpha}}$, then there are two cases: Either α is the successor of a limit ordinal, or it isn't. If it isn't, then for every limit ordinal λ in α, λ^+ is also in α, meaning there are also $\overline{\overline{\alpha}}$ successor ordinals in α. Otherwise, if α is the successor of a limit ordinal, there still are $\overline{\overline{\alpha}}$ successor ordinals in α, as we could just take any successor ordinal less than ω, which replaces the successor ordinal after λ for λ the predecessor of α. Thus, if there are $\overline{\overline{\alpha}}$ limit ordinals in α, there are also $\overline{\overline{\alpha}}$ successor ordinals in α. Hence, in any case, there are $\overline{\overline{\alpha}}$ successor ordinals in α). Now consider some infinite cardinal c. Then c contains c successor ordinals. Then, for each successor ordinal $\alpha^+ < c$, L_α contains some element not in L_α (for example, L_α itself); call this element α'. Then the set of all α' is of size c (because of course no two α' are the same), meaning L_c has at least c elements. Thus, by part (d) of this exercise (and the Bernstein-Schroder Theorem), it follows that $\overline{\overline{L_c}} = c$.

Section 1

Theorem 1.3

To me it seems that we need to have the additional assumption that K can be well founded, so that we even know there is a Mostowski-Shepherdson map over B in the first place. Adding this assumption would not be problematic in using Theorem 1.3 to prove Theorem G, as we already know that L_α for all ordinals α is well founded.

Note that $a' = a$ is due to Theorem 5.1 of Chapter 10.

Section 2

Notes

Clearly C_1 should really be $K_0 = \emptyset$, not $K_0 \neq \emptyset$.

Exercise 2.1

To prove the exercise, we first prove the following useful lemma.

Lemma 0.1. *Suppose two classes A and B are $\in$-isomorphic under a function π. Then, for any subclass C of A, if a class D is definable over C, then $\pi''(D)$ is definable over $\pi''(C)$. In particular, if a formula ϕ (whose constants are $a_1, ..., a_n$) defines D over C, then ψ, which is the result of replacing each a_i in ϕ by $\pi(a_i)$, defines $\pi''(D)$ over $\pi''(C)$.*

Proof. Assume the hypothesis. Consider any subclass C of A. We prove this by induction on the degree of the formulas.

Atomic case: In this case, the formula would have to either be of the form $x \in a$ or $a \in x$, where $a \in C$. Suppose that it is the first case; that is, the formula is of the form $x \in a$ (where $a \in C$). We show that $\pi''(D)$ is definable over $\pi''(C)$ by $x \in \pi(a)$. Well, note that $\pi(a)$ is indeed a member of $\pi''(C)$ (as $a \in C$). On the one hand, suppose that $\pi(b) \in \pi(a)$ holds for some $\pi(b) \in \pi''(C)$ (where $b \in C$). Then, because π is an $\in$-isomorphism, $b \in a$, meaning $b \in D$, and thus $\pi(b) \in \pi''(D)$. On the other hand, suppose that $\pi(b) \in \pi''(D)$ (for some $b \in D$). Then $b \in D$, meaning $b \in a$, and thus, because π is an $\in$-isomorphism, $\pi(b) \in \pi(a)$. Thus $x \in \pi(a)$ defines $\pi''(D)$ over $\pi''(C)$, therefore completing the first case. The second case (when the formula is of the form $a \in x$) is nearly identical.

Negation case: In this case, suppose the result holds for some formula $\phi(x)$, and that $\sim\phi(x)$ defines D over C. Suppose also that the only constants in $\phi(x)$ are $a_1, ..., a_n$. We show that $\sim\psi(x)$, where ψ is the result of replacing each a_i in ϕ by $\pi(a_i)$, defines $\pi''(D)$ over $\pi''(C)$. First note that all constants in $\sim\psi$ are in $\pi''(C)$ (as each constant is of the form $\pi(a_i)$ where $a_i \in C$). On the one hand, suppose that $\sim\psi(\pi(a))$ holds for some $\pi(a) \in \pi''(C)$ (where $a \in C$). By assumption, it follows that $\pi(a) \in \pi''(D)$ (because $\phi(x)$ defines $C - D$, and because $\sim\psi(\pi(a))$ is true, we know that $\psi(\pi(a))$ is false, hence $\pi(a) \in \pi''(C)$ but $\pi(a) \notin \pi''(C) - \pi''(D)$, meaning $\pi(a) \in \pi''(D)$).

On the other hand, if $\pi(a) \in \pi''(D)$ (for some $a \in D$), then $a \in D$, meaning $\sim\psi(\pi(a))$ is true over $\pi''(C)$ (by assumption of ϕ).

Conjunction case: In this case, suppose that the result holds for some formulas ϕ (with constants $a_1, ..., a_n$, and which defines some class K) and ψ (with constants $b_1, ..., b_n$, and which defines some class L), and that the formula $\gamma(x)$: $\phi \wedge \psi$ defines D over C. Let ϕ' and ψ' be the results of replacing the occurences of all a_i in ϕ with $\pi(a_i)$, and the occurences of all b_i in ψ with $\pi(b_i)$, respectively. Then let $\gamma'(x)$ be $\phi' \wedge \psi'$. Then, of course, all constants of $\gamma'(x)$ are in $\pi''(C)$.

On the one hand, suppose that $\gamma'(\pi(a))$ holds for some $\pi(a) \in \pi''(C)$. Then either $\phi'(\pi(a))$ or $\psi'(\pi(a))$ is true over $\pi''(C)$. Suppose that $\phi'(\pi(a))$ is true over $\pi''(C)$ (the case where $\psi'(\pi(a))$ is true over $\pi''(C)$ is identical). Then $\pi(a) \in \pi''(K)$ (by assumption), hence $a \in K$, hence $\phi(a)$ is true over C, hence $\gamma(a)$ is true over C, hence $a \in D$, and therefore $\pi(a) \in \pi''(D)$.

On the other hand, suppose that $\pi(a) \in \pi''(D)$. Then $a \in D$, and hence either $\phi(a)$ or $\psi(a)$ is true over C. Without loss of generality, suppose that $\phi(a)$ is true over C. Then $\phi'(\pi(a))$ is true over $\pi''(C)$ (by assumption), meaning $\gamma'(\pi(a))$ is also true over $\pi''(C)$.

Existential case: In this case, consider some formula of the form $(\exists x)\phi(x, y)$ (with constants $a_1, ..., a_n$) which defines some class D over C. Let $\phi'(x, y)$ be the result of replacing each occurence of a a_i in $\phi(x, y)$ with $\pi(a_i)$. Then, of course, all constants of $(\exists x)\phi'(x, y)$ are in $\pi''(C)$.

On the one hand, suppose that $(\exists x)\phi'(x, \pi(a))$ holds for some $\pi(a) \in \pi''(C)$. Then, for some $\pi(b) \in \pi''(C)$, $\phi'(\pi(b), \pi(a))$ is true over $\pi''(C)$, meaning, by assumption (applied to $\phi(b, x)$), $\phi(b, a)$ is true over C, hence $(\exists x)\phi(x, a)$ is true over C, hence $a \in D$, hence $\pi(a) \in \pi''(D)$.

On the other hand, suppose that $\pi(a) \in \pi''(D)$. Then $a \in D$, hence $\phi(b, a)$ is true over C for some $b \in C$. Then, by assumption (applied to $\phi(b, x)$), it follows that $\phi'(\pi(b), \pi(a))$ is true over $\pi''(C)$, hence $(\exists x)(x, \pi(a))$ is true over $\pi''(C)$. $\square$

Now we can prove Exercise 2.1. That is, we suppose that $(K_0, K_1, ..., K_\alpha, ...)$ $(\alpha < \gamma)$ is a pseudo-construction and that $K = \cup_\alpha < \gamma K_\alpha$, and that π is a Mostowski-Shpepherdson map on K. We prove that

(a) For each $\alpha < \gamma$, $\pi''(K_\alpha) = L_\alpha$.

(b) $\pi''(K) = L_\gamma$

Proof. (a) By transfinite induction.

Least ordinal case: This is given by C_1.

Successor case: Suppose the result holds for some $\alpha < \lambda$ (and that $\alpha^+ < \lambda$), that is, $\pi''(K_\alpha) = L_\alpha$. We must show that $\pi''(K_{\alpha^+}) = L_{\alpha^+} = \mathcal{F}(L_\alpha)$.

On the one hand, suppose that $x \in \pi''(K_{\alpha^+})$. Then $x = \pi(a)$ for some $a \in K_{\alpha^+}$. By C_2, $a \cap K \in \mathcal{F}(K_\alpha)$. Also, because π is a Mostowski-Shepherdson map, $\pi(a) = \pi''(a \cap K)$. By the above lemma, it follows that $\pi''(a \cap K)$ is definable over $\pi''(K_\alpha) = L_\alpha$, meaning $\pi''(a \cap K) \in \mathcal{F}(L_\alpha) = L_{\alpha^+}$, that is, $\pi(a) \in L_{\alpha^+}$. Hence $\pi''(K_{\alpha^+}) \subseteq L_{\alpha^+}$.

On the other hand, suppose that $x \in L_{\alpha^+} = \mathcal{F}(L_\alpha)$. By the above lemma, it follows that $(\pi^{-1})''(x) \in \mathcal{F}(K_\alpha)$. Thus (by C_2), for some $a \in K_{\alpha+1}$, $a \cap K = (\pi^{-1})''(x)$. Then $\pi(a) \in \pi''(K_{\alpha+1})$. But $\pi(a) = \pi''(a \cap K)$ (because π is a Mostowski-Shepherdson mapping) and $\pi''(a \cap K) = \pi''((\pi^{-1})''(x)) = x$, and thus $x \in \pi''(K_{\alpha+1})$. Thus $L_{\alpha+1} \subseteq \pi''(K_{\alpha+1})$.

Hence we have shown that $\pi''(K_{\alpha^+} = L_{\alpha^+}$.

Limit ordinal case: This is obvious (by C_3).

(b) Obvious (using part (a) of this exercise). ☐

Exercise 2.2

Suppose K and M are sets, and K is bound up with M. We prove that $\mathcal{F}(K \cap M)$ is the set of all elements $a \cap K$ such that $a \in (K \cap \mathcal{F}(M))$.

Proof. On the one hand, suppose that $x \in \mathcal{F}(K \cap M)$. Because $K \cap M$ is Henkin closed with respct to M (by B_1), it follows that $x \in \mathcal{F}(M)$; more specifically, x is definable over M by a formula whose constants are all in $K \cap M$ (this is just the same formula that defines x over $K \cap M$). By B_2, it follows that $x \in K$. Thus $x \in (K \cap \mathcal{F}(M))$. But also $x = x \cap K$ (because all elements of x are in $K \cap M$, hence in K). Hence there is some a (namely x itself) such that $x = a \cap K$ and $a \in (K \cap \mathcal{F}(M))$.

On the other hand, suppose that $x = a \cap K$ where $a \in (K \cap \mathcal{F}(M))$. By B_2, it follows that a is definable over M by a formula ϕ whose constants are all in $K \cap M$. Because $K \cap M$ is Henkin closed with respect to M, it follows that ϕ defines $a \cap K$ over $K \cap M$, meaning $a \cap K \in \mathcal{F}(K \cap M)$.

Thus we have shown that $\mathcal{F}(K \cap M)$ is the set of all elements $a \cap K$ such that $a \in (K \cap \mathcal{F}(M))$.

☐

Exercise 2.3

We are to show that the γ-tuple $(K_0, ..., K_\alpha, ...)$ $(\alpha < \gamma)$ is a pseudo-construction (of K).

C_1: We show that $K_0 = \emptyset$. Well, $K_0 = K \cap L_{\overline{0}}$. Now, $\overline{0}$ is the least ordinal in $o(K)$, i.e., the lowest order of any element in K. What this means is that $K \cap L_{\overline{0}} = \emptyset$, and thus $K_0 = \emptyset$.

C_2: Consider any $\alpha < \gamma$. Then K is bound up with $L_{\overline{\alpha}}$. Also, $K_{\alpha+1} = K \cap L_{\overline{\alpha+1}}$, $K_\alpha = K \cap L_{\overline{\alpha}}$. Note that $L_{\overline{\alpha+1}} = \mathcal{F}(L_{\overline{\alpha}})$ (because $\overline{\alpha}+1 \in o(K)$, meaning $L_{\overline{\alpha+1}} = L_{\overline{\alpha}+1} = \mathcal{F}(L_{\overline{\alpha}})$). Then $\mathcal{F}(K \cap L_{\overline{\alpha}})$ is the set of all elements $a \cap K$

such that $a \in (K \cap L_{\overline{\alpha+1}})$ (by Exercise 2.2). But $\mathcal{F}(K \cap L_{\overline{\alpha}}) = \mathcal{F}(K_\alpha)$, and $K \cap L_{\overline{\alpha+1}} = K_{\alpha+1}$. Thus $\mathcal{F}(K_\alpha)$ is the set of all elements $a \cap K$ where $a \in K_{\alpha+1}$.

C_3: Consider some limit ordinal $\delta < \gamma$. Then $K_\delta = K \cap L_{\overline{\delta}}$. We are to show that $K_\delta = \cup_{\alpha<\delta} K_\alpha$.

On the one hand, suppose that $x \in \cup_{\alpha<\delta} K_\alpha$. Then $x \in K_\alpha$ for some $\alpha < \delta$. Now, $K_\alpha = K \cap L_{\overline{\alpha}}$. But $\overline{\alpha} < \overline{\delta}$. Hence $x \in L_{\overline{\delta}}$, and thus $x \in K_\delta$ (as $K_\delta = K \cap L_{\overline{\delta}}$).

On the other hand, suppose that $x \in K_\delta$. Now, $\overline{\delta}$ must be a limit ordinal. Hence, because $x \in L_{\overline{\delta}}$, it follows that $x \in L_\alpha$ for some $\alpha < \overline{\delta}$. But, because also $x \in K$, it follows that there must be some $\beta \in o(K)$, where $\beta < \delta$, such that $x \in L_{\overline{\beta}}$. Hence $x \in K \cap L_{\overline{\beta}}$, meaning $x \in K_\beta$, for some $\beta < \delta$.